

Historical Stock Analysis

Everything you need to explain the current 10-slide PPT without reading the full handbook.

Central question

Does the diversification visible in a global stock dataset survive closer inspection across technology groups and crisis periods?

Answer in one sentence

Diversification looks thinner when selected chip/AI stocks dominate technology growth and when regional returns move together during sudden panic, especially in 2020.

This is a historical data-visualization project. It describes patterns; it is not a trading system, a forecast, or proof that one event caused another.

Dataset	Coverage	Quality
230,111 rows x 8 columns	49 tickers, 2006-01-02 to 2026-02-20	0 missing, 0 duplicate Date-Ticker pairs
One row = one ticker-day	Six market labels, uneven coverage	1,706 zero-volume rows

- H1 asks whether the post-2023 technology rally was concentrated in selected chip/AI stocks.
- H2 asks whether regional correlations rose during crises.
- H3 asks whether volatility and volume rose during panic periods.

Remember

The original Kaggle page has been removed by its uploader, and the data stop on 2026-02-20. Do not call the last observation current.

The five ideas behind every chart

If these are clear, the presentation becomes straightforward.

1. Adjusted Close

Use this price for growth and returns because it accounts for historical splits and dividends.

2. Daily return

Return = today's Adj Close / previous Adj Close - 1. It is a percentage change, so it removes currency units.

3. Index-to-100

Indexed value = $100 \times \text{current Adj Close} / \text{starting Adj Close}$. Ending at 649 means a 549% gain, not 649%.

4. Correlation

Ranges from -1 to +1. A higher positive value means returns move together more, so they offer less diversification from one another.

5. Volatility and volume

Volatility is movement size. The monthly chart uses absolute returns so gains and losses do not cancel. Volume is share count and is only supporting evidence because splits and company size affect it.

Why not raw prices? The tickers use different currencies and per-share scales. Compare returns or rebase each series to 100 instead.

Exactly what was compared

Keep the selected groups and exact time windows visible when explaining the results.

Group	Tickers
Chip / AI	NVDA, AVGO, TSM, ASML, AMD, MU, LRCX
Broader tech	AAPL, MSFT, GOOGL, AMZN, META, NFLX, ORCL, CSCO
One-stock regions	JPM, 005930.KS, 0700.HK, NOVN.SW, MC.PA
Widened regions	US 3; Korea 2; Hong Kong 3; Switzerland 2; Paris still 1

Period	Exact dates	Meaning
Calm	2015-01-01 to 2017-12-31	Reference baseline
2008 crisis	2008-09-01 to 2009-03-31	Global banking shock
2020 crash	2020-02-15 to 2020-04-30	Sudden pandemic selloff
2022 drawdown	2022-01-01 to 2022-12-31	Slower inflation/rate shock

H1

Selected chip/AI stocks will outperform selected broader tech after 2023.

H2

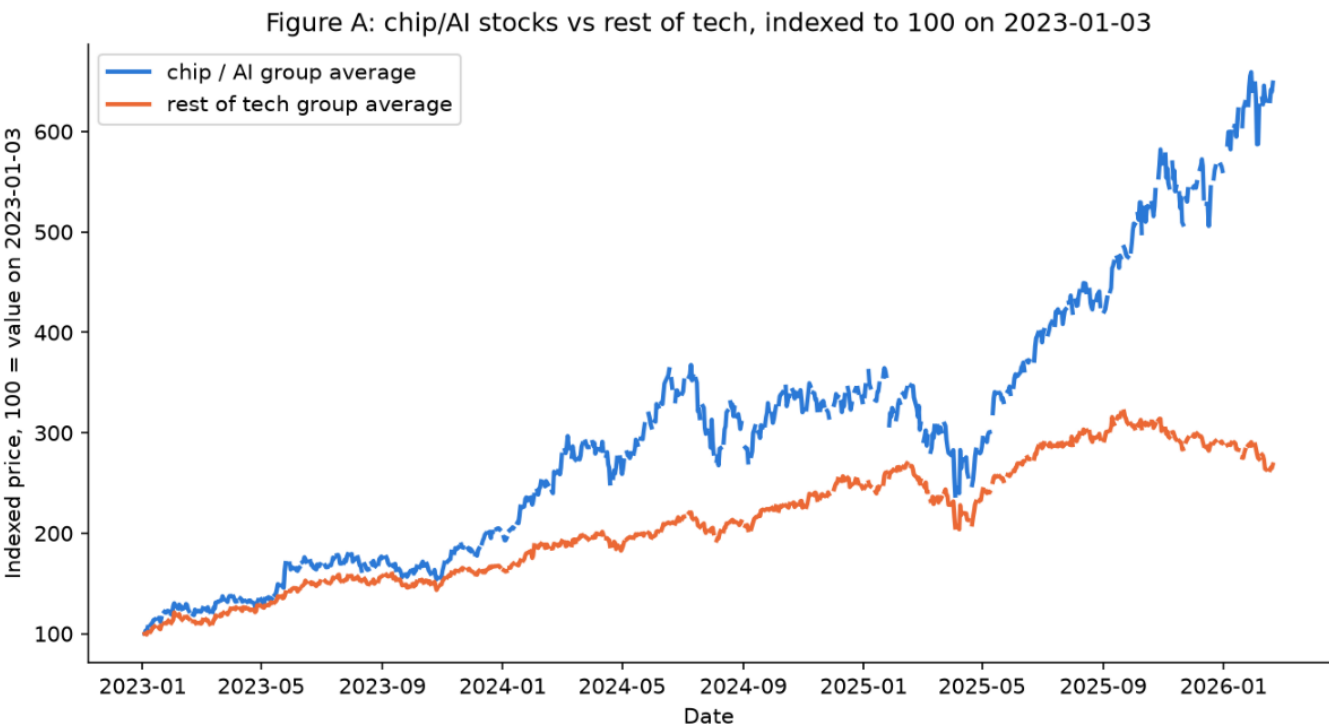
Regional correlations will be higher in crises than in calm years.

H3

Volatility and volume will rise during panic periods.

Selected chip/AI stocks pulled decisively ahead

Slide 5: indexed group averages from 2023-01-03 through 2026-02-20.



Source: data/stock_data.csv, 2023-01-03 to 2026-02-20. Chip/AI group: NVDA, AVGO, TSM, ASML, AMD, MU, LRCX.
Rest of tech group: AAPL, MSFT, GOOGL, AMZN, META, NFLX, ORCL, CSCO. Each group line is a simple average of the indexed prices.

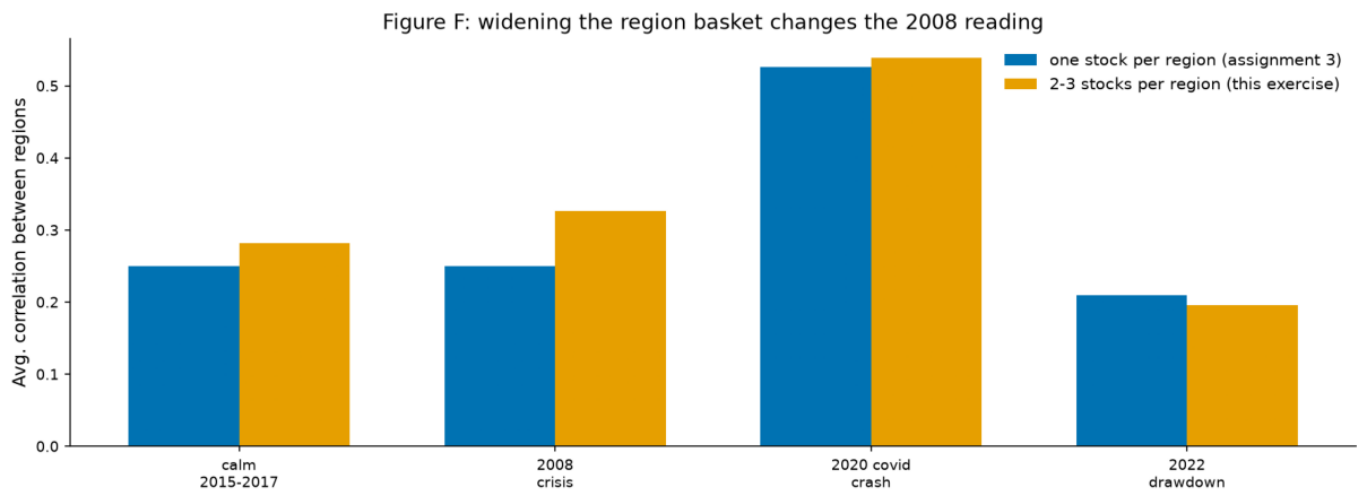
Measure	Chip / AI	Broader tech
Cumulative gain	about +549%	about +168%
Mean daily return	0.26%	0.14%
95% confidence interval	0.19% to 0.34%	0.09% to 0.19%

Statistical check and conclusion

Welch $t = 2.69$ and $p = 0.0073$. H1 is supported for these selected groups, not every AI-linked company, and the test does not prove that AI caused the gap.

Regional diversification weakened most in 2020

Slide 6 uses widened regional baskets where the dataset allows.



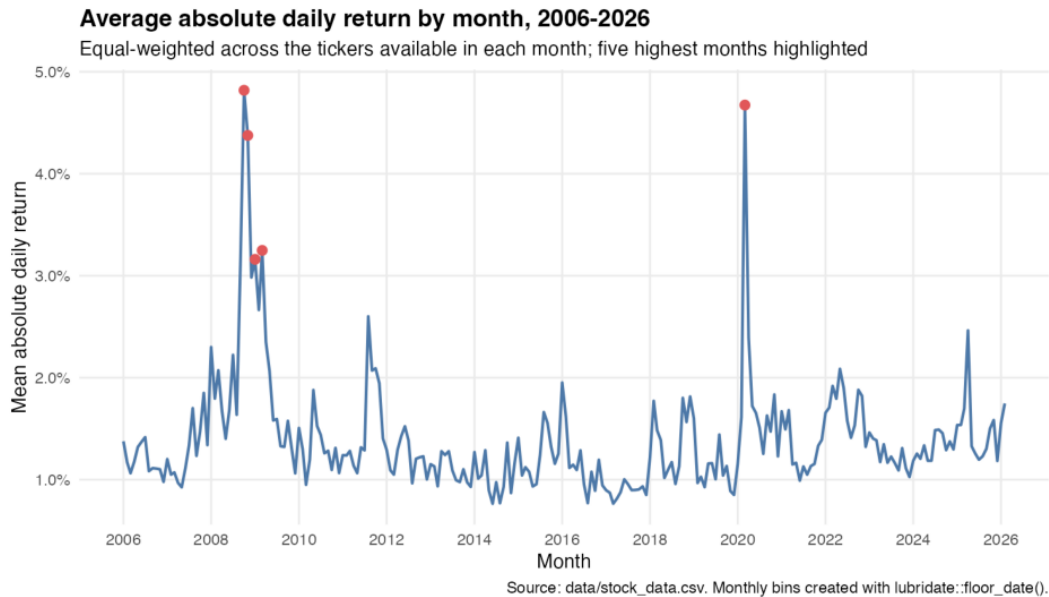
Period	One stock	Widened basket	Reading
Calm	0.250	0.282	Baseline
2008	0.250	0.327	Directionally higher
2020	0.529	0.539	Clear maximum
2022	0.209	0.196	Below calm

Conclusion with limitation

H2 is strongly supported for 2020 and directionally supported for 2008; 2022 did not show the same synchronized pattern. Paris still has one stock, while Korea and Switzerland remain sector-narrow, so this is qualified evidence rather than a universal law.

Panic periods spiked; the redesign made growth readable

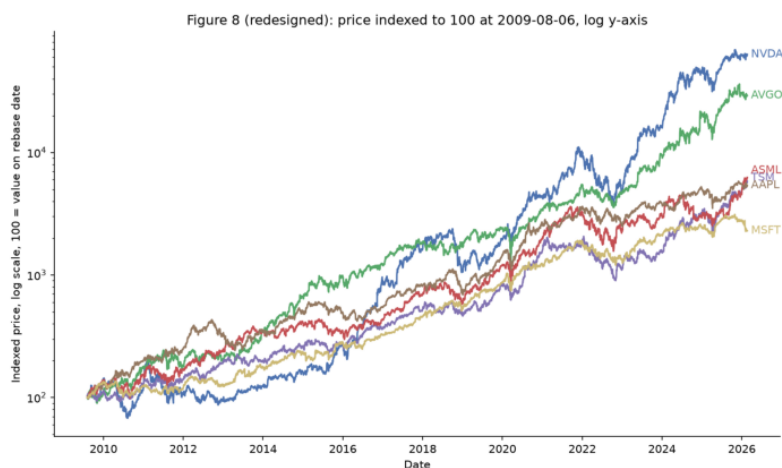
Slides 7 and 8: independent R volatility evidence, then a visualization lesson.



H3 headline

October 2008 had the highest monthly mean absolute daily return at 4.82%; March 2020 was second at 4.67%. One-stock volume was about 2.7x calm in 2008, 1.7x in 2020, and 0.9x in 2022.

This supports H3: panic periods in this dataset show strong volatility and volume spikes. Do not say that volume caused the crashes.



Design lesson: the original linear chart compressed four lines. A log y-axis makes equal vertical distance represent equal proportional change; separated direct labels reduce clutter. The data did not change.

What to say on each of the 10 slides

One job per slide. Use the headline, one explanation, and one limitation where needed.

Slide	Speaker	Core line
1	Mirthula	Open: this is diversification analysis, not forecasting.
2	Mirthula	Dataset, one ticker-day, Python/R, returns instead of raw prices.
3	Mirthula	State H1, H2, H3 before any evidence.
4	Mirthula	Define groups, periods, correlation, volatility, and volume.
5	Karthikeyan	H1: +549% vs +168%; $p = 0.0073$; selected groups only.
6	Karthikeyan	H2: 2020 correlation 0.539; qualify 2008 and regional coverage.
7	Karthikeyan	H3: Oct 2008 4.82%, Mar 2020 4.67%; volume supports.
8	Lakshmi	Linear vs log: proportional comparison and clearer labels.
9	Lakshmi	Resolve question: diversification is thinner, within limits.
10	Lakshmi	Next: filterable Power BI/Tableau dashboard with visible assumptions.

Handovers

Mirthula: 'Now Karthikeyan will show whether the hypotheses are supported.' Karthikeyan:
'Lakshmi will show how the design improved and close with next steps.'

Final sentence

'Success means the user can reproduce each headline and see where the evidence becomes uncertain.'

Numbers, boundaries, and likely questions

Read this page immediately before presenting.

Remember	Value
Dataset	230,111 rows; 49 tickers; 2006-01-02 to 2026-02-20
H1	+549% vs +168%; 0.26% vs 0.14%; $p = 0.0073$
H2 widened	0.282 calm; 0.327 in 2008; 0.539 in 2020; 0.196 in 2022
H3	4.82% Oct 2008; 4.67% Mar 2020; volume 2.7x / 1.7x / 0.9x

Why not raw prices?

Different currencies and per-share scales. Returns and index-to-100 are comparable.

What does $p = 0.0073$ mean?

The gap is unlikely under an equal-means null model. It is not the probability that H1 is true.

Why widen regions?

One company can carry company-specific news. Several stocks give a broader regional signal.

What can we claim?

Patterns for selected groups and periods. Correlation does not prove causality, and we cannot forecast or state a universal law.

Memory line: 549-168 for H1; 282-327-539-196 for H2; 4.82-4.67 for H3.